

Holonomic Rank, Exact Overlaps, and Non-P-Recursiveness in the Fabius–Rvachev System

Finite sinc-product ODEs, signed-overlap entropy, Thue–Morse spectra,
and the transition to infinite differential rank

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Abstract

The Fabius–Rvachev corpus already contains a remarkably broad theory of dyadic arithmetic, Thue–Morse identities, infinite sinc products, endpoint and inverse endpoint asymptotics, Lambert– W carriers, q -analogues, Legendre expansions, and Bell–Bernoulli calculi. This report develops a complementary axis that was not found in the audited LaTeX corpus: *holonomic complexity*.

For the finite geometric sinc product

$$\Phi_{q,N}(z) = \prod_{j=1}^N \operatorname{sinc}(q^j z), \quad 0 < q < 1,$$

we prove that the minimal order over $\mathbb{C}(z)$ of a homogeneous linear differential equation for $\Phi_{q,N}$ is exactly the number of nonzero atoms in a signed finite Bernoulli convolution. The same number is the support size of the N th distributional derivative of the corresponding box-spline density. This gives a rank–knot duality: exact overlaps of signed geometric sums are precisely holonomic-rank defects. The maximal rank is 2^N if and only if q is not a zero of a nonzero polynomial of degree below N with coefficients in $\{-1, 0, 1\}$. The ranks are submultiplicative, so their exponential rate exists; it equals 2 off the exact-overlap set and is strictly smaller than 2 at every exact-overlap parameter.

In the dyadic case the complete minimal operator is explicit. Set

$$Y_N(z) = z^N \Phi_{1/2,N}(z).$$

Then

$$\prod_{\ell=1}^{2^N-1} \left(D^2 + \frac{(2\ell-1)^2}{4^N} \right) Y_N = 0,$$

and no lower-order rational-coefficient equation exists. Its roots form the odd dyadic frequency lattice, its exponential coefficients are the Thue–Morse signs, and the operator admits a two-sided Pochhammer factorization. We give all-order coefficient formulae in complete Bell polynomials whose power sums are Bernoulli-polynomial values, as well as a Barnes- G discriminant formula.

For the infinite Rvachev transform $\Phi_b(z) = \prod_{j \geq 1} \operatorname{sinc}(z/b^j)$, $b \in \{2, 3, \dots\}$, the familiar unbounded zero multiplicities imply a sharp opposite conclusion: Φ_b is not D-finite. In the dyadic case its logarithmic derivative escapes every finite trigonometric rational field $\mathbb{C}(z, e^{iz/2^M})$, although each finite truncation belongs to one. Consequently the moment sequence, cumulant sequence, and reciprocal-moment coefficients of the Rvachev law are not P-recursive. A separate endpoint argument proves that the exact rational sequence $F(2^{-n})$, and more generally $F(c2^{-n})$ and fixed-order derivative samples, are not P-recursive; their ordinary generating functions are non-D-finite entire functions of order zero.

All theorem and corollary environments below are proved in the text. Exact computations are labelled separately. The golden-overlap rank recurrence and the proposed differential-transcendence and inverse-sampling statements are explicitly presented as conjectures.

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1 Scope, corpus audit, and status of the results

1.1 Repository snapshot and exclusion boundaries

The source audit used the repository snapshot

VladimirReshetnikov/ProveIt at commit 03b2f61889674f7d64ac86d3233236f5fa7ce660.

A recursive code-index query for `path:Analysis/FabiusFunction/docs extension:tex` returned 131 LaTeX paths. The global draft manifest, the canonical primary exposition, and the nearest recent frontier compilations were read directly; corpus-wide phrase searches were then used to test candidate claims against all indexed sources. The machine-readable audit notes are included as `CORPUS_AUDIT.md` in the accompanying archive.

Several tempting directions were rejected because the repository already contains them in substantial form. In particular, this report does *not* claim novelty for the following ingredients:

- (i) the zero set of the dyadic infinite sinc product, the multiplicity $1 + \nu_2(m)$ at $2\pi m$, its spectral zeta function, or the associated digit-counting laws;
- (ii) Fourier-decay envelopes, shell-dependent gauges, and log-periodic spectral asymptotics;
- (iii) the corrected all-orders endpoint and inverse-endpoint expansions, including Lambert– W and periodic Gamma–zeta fluctuations;
- (iv) finite and infinite q -Pochhammer, Gaussian-binomial, inverse- q , cyclotomic, Legendre/Lagrange, dyadic-comb, Stein, zero-bias, total-positivity, and noncommutative layers.

The first item will be reproved briefly because it is the shortest input to a new non-D-finiteness theorem. The novelty is in what is deduced from it, not in the divisor itself.

Exact searches found no use of the terms *D-finite*, *holonomic rank*, *P-recursive*, *minimal differential equation*, or *exponential polynomial* in the documentation tree. Targeted external searches likewise did not locate the particular rank–overlap dictionary and its consequences developed here. This is evidence of corpus-relative novelty, not a claim of absolute historical priority.

1.2 Three levels of assertion

The report keeps the following categories disjoint.

PROVED A complete argument is given here. Some proofs use standard facts about linear ODEs, D-finite series, or distributions, which are stated in the form needed.

EXACT COMPUTATION

Exact rational or exact quadratic-field arithmetic has verified the assertion through the stated finite range. Such a statement is not promoted to a theorem merely because the range is large.

CONJECTURAL A plausible infinite statement supported by structure and/or computation, but not proved here.

1.3 Main new theorem families

The proved contributions can be read in five layers.

1. **Finite rank and splines.** The minimal differential order of $\Phi_{q,N}$ is a signed-frequency support size, also equal to the number of surviving knots in the highest distributional derivative of a finite geometric box spline.
2. **Arithmetic overlap detection.** Full rank is equivalent to the absence of a $\{-1, 0, 1\}$ polynomial relation at q . A submultiplicative rank entropy detects exact overlaps exactly.

3. **Dyadic closed form.** The minimal operator has an odd-lattice factorization, a Pochhammer form, Bell–Bernoulli coefficient formulae, and a Barnes- G discriminant.
4. **Infinite non-holonomicity.** The infinite transform is not D-finite; its logarithmic derivative lies in no finite dyadic trigonometric rational field. Moment, cumulant, and reciprocal coefficient sequences are not P-recursive.
5. **Endpoint sample obstruction.** The deep dyadic samples $F(c2^{-n})$ and their fixed derivative analogues satisfy no polynomial- coefficient recurrence. Their ordinary generating functions are entire of order zero but non-D-finite.

2 Normalizations and background

2.1 The bounded Fabius function and the up function

We use the bounded Fabius function $F : \mathbb{R} \rightarrow [0, 1]$, clamped to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$, with

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}, \quad (2.1)$$

and $F(x) + F(1 - x) = 1$ on $[0, 1]$. It is strictly increasing on $[0, 1]$, C^∞ , and flat at the endpoints. Rvachev's even up function is the compactly supported density on $[-1, 1]$ related by

$$\text{up}(t) = F(t + 1) \quad (-1 \leq t \leq 0), \quad \text{up}(t) = F(1 - t) \quad (0 \leq t \leq 1). \quad (2.2)$$

These normalizations agree with the primary exposition in the repository and with Arias de Reyna's account [2, 3].

The probabilistic representation most useful here is

$$X = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1], \quad (2.3)$$

whose density is up in the chosen scaling. Its characteristic function is

$$\Phi(z) = \mathbb{E} e^{izX} = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{z}{2^j}\right), \quad \text{sinc } w := \begin{cases} \sin w/w, & w \neq 0, \\ 1, & w = 0. \end{cases} \quad (2.4)$$

The product converges locally uniformly on $\mathbb{C}$ and hence defines an entire function.

2.2 A geometric finite family

For $0 < q < 1$ and $N \geq 1$, let

$$X_{q,N} = \sum_{j=1}^N q^j U_j, \quad \Phi_{q,N}(z) = \mathbb{E} e^{izX_{q,N}} = \prod_{j=1}^N \text{sinc}(q^j z). \quad (2.5)$$

Let $u_{q,N}$ be the density of $X_{q,N}$. It is the convolution of the normalized boxes

$$h_a(x) = \frac{1}{2a} \mathbf{1}_{[-a,a]}(x), \quad u_{q,N} = h_q * h_{q^2} * \cdots * h_{q^N}. \quad (2.6)$$

Thus $u_{q,N}$ is a compactly supported spline of degree at most $N - 1$. The dyadic densities converge to up in the standard senses.

A second random object, involving Rademacher rather than uniform digits, will encode the finite differential complexity:

$$W_{q,N} = \sum_{j=1}^N \varepsilon_j q^j, \quad \mathbb{P}(\varepsilon_j = 1) = \mathbb{P}(\varepsilon_j = -1) = \frac{1}{2}. \quad (2.7)$$

Exact coincidences among the values of $W_{q,N}$ are the finite exact overlaps of the associated Bernoulli convolution. The classical theory of Bernoulli convolutions studies far subtler measure and dimension questions [11, 8]; here only the finite signed-sum arithmetic is needed.

2.3 D-finiteness, P-recursiveness, and rank

Definition 2.1 (D-finite function). A germ f is *D-finite over* $\mathbb{C}(z)$ if there are polynomials $p_0, \dots, p_r \in \mathbb{C}[z]$, with $p_r \neq 0$, such that

$$p_r(z)f^{(r)}(z) + \dots + p_1(z)f'(z) + p_0(z)f(z) = 0. \quad (2.8)$$

Equivalently, the $\mathbb{C}(z)$ -vector space spanned by $f, f', f'', \dots$ is finite-dimensional. Its dimension, or equivalently the least possible order r , will be called the *differential rank* of f .

Definition 2.2 (P-recursive sequence). A sequence $(a_n)_{n \geq 0}$ is *P-recursive* if there are polynomials $p_0(n), \dots, p_r(n)$, not all zero, for which

$$\sum_{j=0}^r p_j(n)a_{n+j} = 0 \quad (2.9)$$

for every sufficiently large integer n .

Stanley's fundamental dictionary says that (a_n) is P-recursive if and only if its ordinary generating series $\sum a_n z^n$ is D-finite [12]. The analogous statement for an exponential generating function follows because multiplication or division by $n!$ preserves P-recursiveness after clearing the shifted factorials in a recurrence.

Two elementary ODE facts will be used repeatedly.

Lemma 2.3 (Ordinary-point multiplicity bound). *Let $f \not\equiv 0$ satisfy an order- r equation of the form (2.8). If $p_r(z_0) \neq 0$, then f cannot have a zero of multiplicity at least r at z_0 .*

Proof. At an ordinary point z_0 , the values $f(z_0), \dots, f^{(r-1)}(z_0)$ determine the local solution uniquely. If all are zero, the unique solution is the zero solution. A zero of multiplicity at least r would give exactly those zero initial data. $\square$

Lemma 2.4 (Finite singular locus). *A D-finite germ can be analytically continued along every path avoiding the finite zero set of the leading polynomial p_r . In particular, a meromorphic function with infinitely many finite-plane poles is not D-finite.*

Proof. On any simply connected domain where p_r does not vanish, (2.8) is a nonsingular first-order system for $(f, f', \dots, f^{(r-1)})$. Standard analytic existence and continuation for linear systems gives the assertion. $\square$

3 Finite signed-frequency calculus and exact differential rank

3.1 The exponential polynomial behind a finite sinc product

Set

$$Y_{q,N}(z) := z^N \Phi_{q,N}(z). \quad (3.1)$$

Since $\sin x = (2i)^{-1} \sum_{\varepsilon=\pm 1} \varepsilon e^{i\varepsilon x}$,

$$Y_{q,N}(z) = \frac{q^{-N(N+1)/2}}{(2i)^N} \sum_{\varepsilon \in \{\pm 1\}^N} \left(\prod_{j=1}^N \varepsilon_j \right) e^{iz\omega(\varepsilon)}, \quad \omega(\varepsilon) := \sum_{j=1}^N \varepsilon_j q^j. \quad (3.2)$$

The prefactor is irrelevant to every rank question.

When several sign vectors yield the same frequency, their parity signs must be aggregated. Define

$$c_{q,N}(\omega) := \sum_{\substack{\varepsilon \in \{\pm 1\}^N \\ \omega(\varepsilon) = \omega}} \prod_{j=1}^N \varepsilon_j, \quad \Omega_{q,N}^* := \{\omega : c_{q,N}(\omega) \neq 0\}, \quad r_N(q) := |\Omega_{q,N}^*|. \quad (3.3)$$

The distinction between the set of distinct signed sums and $\Omega_{q,N}^*$ is essential: different words can collide and then cancel completely.

3.2 A minimal-rank lemma for exponential polynomials

Lemma 3.1 (Polynomial-exponential independence). *If $\lambda_1, \dots, \lambda_h$ are distinct complex numbers, then $e^{\lambda_1 z}, \dots, e^{\lambda_h z}$ are linearly independent over $\mathbb{C}(z)$.*

Proof. After multiplying by a common denominator, it is enough to exclude a nontrivial relation

$$\sum_{j=1}^h p_j(z) e^{\lambda_j z} = 0, \quad p_j \in \mathbb{C}[z]. \quad (3.4)$$

Proceed by induction on the number of nonzero terms. Choose a nonzero last term and take $M > \deg p_h$. Applying $(D - \lambda_h)^M$ kills $p_h(z) e^{\lambda_h z}$. For $j < h$ it produces

$$e^{\lambda_j z} (D + \lambda_j - \lambda_h)^M p_j(z).$$

Because $\lambda_j - \lambda_h \neq 0$, the operator $D + \lambda_j - \lambda_h$ is injective on $\mathbb{C}[z]$: its action preserves the degree and multiplies the leading coefficient by a nonzero scalar. Hence no remaining nonzero term is killed. We obtain a shorter relation, contradicting the induction hypothesis. $\square$

Lemma 3.2 (Exact rank of an exponential polynomial). *Let*

$$H(z) = \sum_{j=1}^h a_j e^{\lambda_j z}, \quad a_j \neq 0,$$

with distinct λ_j . Then the differential rank of H over $\mathbb{C}(z)$ is exactly h , and its minimal monic constant-coefficient annihilator is

$$\prod_{j=1}^h (D - \lambda_j). \quad (3.5)$$

Proof. The displayed operator annihilates every summand, so the rank is at most h . If $H, H', \dots, H^{(h-1)}$ were dependent over $\mathbb{C}(z)$, then for some rational functions $b_0, \dots, b_{h-1}$,

$$0 = \sum_{k=0}^{h-1} b_k(z) H^{(k)}(z) = \sum_{j=1}^h a_j \left(\sum_{k=0}^{h-1} b_k(z) \lambda_j^k \right) e^{\lambda_j z}.$$

By [Theorem 3.1](#), every parenthesis vanishes. The polynomial in t given by $\sum_{k=0}^{h-1} b_k(z) t^k$ has the h distinct roots λ_j but degree below h , so all b_k vanish. Thus the first h derivatives are independent and the rank is at least h . $\square$

Multiplication by a nonzero rational function does not change differential rank: the derivatives of rf and those of f are related at each finite level by an invertible triangular matrix over $\mathbb{C}(z)$ with diagonal r . Applying this observation to the gauge in (3.1) gives the central finite theorem.

Theorem 3.3 (Exact finite holonomic rank). *For every $0 < q < 1$ and $N \geq 1$, the differential rank of $\Phi_{q,N}$ over $\mathbb{C}(z)$ is exactly*

$$\boxed{\text{rank}_{\mathbb{C}(z)} \Phi_{q,N} = r_N(q) = |\Omega_{q,N}^*|}. \quad (3.6)$$

A minimal constant-coefficient annihilator of $Y_{q,N} = z^N \Phi_{q,N}$ is

$$L_{q,N}(D) = \prod_{\omega \in \Omega_{q,N}^*} (D - i\omega), \quad (3.7)$$

and a minimal rational-coefficient annihilator of $\Phi_{q,N}$ is the gauge conjugate

$$\tilde{L}_{q,N} = z^{-N} L_{q,N}(D) z^N. \quad (3.8)$$

Proof. After equal frequencies in (3.2) are collected, $Y_{q,N}$ is a sum of exactly $r_N(q)$ exponentials with distinct exponents and nonzero coefficients. Apply Theorem 3.2, then use invariance under the rational gauge z^{-N} . $\square$

Remark 3.4 (What the theorem computes). The order is not merely bounded by 2^N ; it is determined exactly by a finite signed-convolution problem. It may be computed with integer arithmetic in the group ring generated by $q, q^2, \dots, q^N$. At an algebraic parameter one can work modulo its minimal polynomial, with no floating-point comparison.

3.3 Rank–knot duality in physical space

The same signed support occurs without Fourier analysis. In the sense of distributions,

$$h'_a = \frac{\delta_{-a} - \delta_a}{2a}. \quad (3.9)$$

Differentiating one factor in every convolution in (2.6) gives the following identity.

Theorem 3.5 (Highest-derivative atomic measure). *The N th distributional derivative of the finite density is*

$$u_{q,N}^{(N)} = \frac{(-1)^N}{2^N q^{N(N+1)/2}} \sum_{\omega \in \Omega_{q,N}^*} c_{q,N}(\omega) \delta_\omega. \quad (3.10)$$

Consequently,

$$\boxed{\text{rank}_{\mathbb{C}(z)} \Phi_{q,N} = \# \text{supp}(u_{q,N}^{(N)})}. \quad (3.11)$$

A signed frequency survives in the minimal differential equation if and only if the same point survives as a knot in the highest derivative of the finite box spline.

Proof. By (3.9), the derivative of the j th box contributes $+\delta_{-q^j}/(2q^j)$ or $-\delta_{q^j}/(2q^j)$. Convolving the N two-point measures gives

$$u_{q,N}^{(N)} = \frac{1}{2^N q^{1+2+\dots+N}} \sum_{\epsilon \in \{\pm 1\}^N} (-1)^N \left(\prod_j \epsilon_j \right) \delta_{\sum_j \epsilon_j q^j}.$$

Collect equal atoms. The coefficient at ω is the scalar in (3.10); its support is therefore $\Omega_{q,N}^*$. Now apply Theorem 3.3. $\square$

Corollary 3.6 (Local spline simplification at a cancellation). *If $c_{q,N}(\omega_0) = 0$, then the nominal knot ω_0 does not create a jump in $u_{q,N}^{(N-1)}$. The two polynomial pieces adjacent to that nominal knot therefore join with one more order of compatibility than the unaggregated word list predicts.*

Proof. The jump of a locally integrable function at ω_0 is exactly the coefficient of δ_{ω_0} in its distributional derivative. Apply this to $u_{q,N}^{(N-1)}$ and use (3.10). $\square$

This is the promised rank–knot duality. The finite Bernoulli-convolution language, the exponential-polynomial language, and the spline-knot language are three representations of the same integer $r_N(q)$.

4 Exact overlaps, separation, and holonomic entropy

4.1 The restricted-coefficient collision polynomial

Let

$$\mathcal{L}_d := \left\{ \sum_{k=0}^d a_k x^k : a_k \in \{-1, 0, 1\} \right\} \subset \mathbb{Z}[x]. \quad (4.1)$$

We include the zero polynomial in $\mathcal{L}_d$ but exclude it whenever roots are discussed. These height-one restricted-coefficient polynomials occur in Littlewood-type approximation problems [6, 5].

Theorem 4.1 (Full-rank criterion). *For $0 < q < 1$ and $N \geq 1$, the following statements are equivalent.*

- (a) $r_N(q) = 2^N$.
- (b) The 2^N signed sums $\sum_{j=1}^N \varepsilon_j q^j$ are pairwise distinct.
- (c) $P(q) \neq 0$ for every nonzero $P \in \mathcal{L}_{N-1}$.

Thus the rank-defect set at level N is a finite algebraic set: it is exactly the set of roots in $(0, 1)$ of nonzero $\{-1, 0, 1\}$ polynomials of degree below N .

Proof. If all signed sums are distinct, every aggregate coefficient is a single number ± 1 , so $r_N(q) = 2^N$. If two sums collide, there are fewer than 2^N distinct frequencies, hence certainly fewer than 2^N nonzero aggregate frequencies. This proves the equivalence of (a) and (b).

For two sign words $\varepsilon, \varepsilon'$, write $a_{j-1} = (\varepsilon_j - \varepsilon'_j)/2 \in \{-1, 0, 1\}$. Then

$$\omega(\varepsilon) - \omega(\varepsilon') = 2q \sum_{k=0}^{N-1} a_k q^k. \quad (4.2)$$

Every nonzero coefficient word $(a_0, \dots, a_{N-1})$ can conversely be realized as half the difference of two sign words: use $(+1, -1)$ when $a_k = 1$, $(-1, +1)$ when $a_k = -1$, and equal signs when $a_k = 0$. Therefore a collision occurs exactly when a nonzero member of $\mathcal{L}_{N-1}$ vanishes at q . $\square$

Corollary 4.2 (Large collision-free classes). *The full rank $r_N(q) = 2^N$ holds for every N in each of the following cases:*

- (i) q is transcendental;
- (ii) $q \in \mathbb{Q} \cap (0, 1)$;
- (iii) $0 < q \leq \frac{1}{2}$.

Proof. A transcendental number is not a root of a nonzero integer polynomial. If a rational $q \in (0, 1)$ were a root of a nonzero $P \in \mathcal{L}_d$, remove initial and terminal zero coefficients. The resulting polynomial has constant and leading coefficients ± 1 ; the rational-root theorem forces $q = \pm 1$, a contradiction.

For $q \leq 1/2$, compare two sign words and let k be their first differing position. The leading discrepancy has magnitude $2q^k$, whereas the finite tail has magnitude strictly below $2 \sum_{j>k} q^j \leq 2q^k$; at $q = 1/2$ strictness follows because the tail is finite. Hence no collision is possible. $\square$

Remark 4.3. The rational corollary is worth isolating. Parameters such as $q = 2/3$ can produce extremely close frequencies, but never an exact finite collision. In contrast, an algebraic irrational such as the inverse golden ratio can produce a defect at level three.

4.2 A sharp separation formula below the overlap threshold

Define the minimum signed-frequency spacing

$$\Delta_N(q) := \min_{\varepsilon \neq \varepsilon'} |\omega(\varepsilon) - \omega(\varepsilon')|. \quad (4.3)$$

It is zero exactly at a level- N overlap parameter. Equation (4.2) immediately gives

$$\Delta_N(q) = 2q \min_{\substack{P \in \mathcal{L}_{N-1} \\ P \neq 0}} |P(q)|. \quad (4.4)$$

Thus near-collisions of frequencies are precisely small values of restricted-coefficient polynomials.

Theorem 4.4 (Exact separation for $q \leq 1/2$). *For $0 < q \leq 1/2$,*

$$\boxed{\Delta_N(q) = 2q^N}. \quad (4.5)$$

Proof. Changing only the last sign produces a pair of sums at distance $2q^N$, so $\Delta_N(q) \leq 2q^N$. For the reverse inequality, let k be the first position at which two words differ and put $d = N - k$. The leading difference has magnitude $2q^k$. Even if all later differences oppose it, the remaining magnitude is at least

$$2q^k \left(1 - \sum_{r=1}^d q^r \right) = 2q^k \frac{1 - 2q + q^{d+1}}{1 - q}.$$

The bracket is at least q^d , because

$$\frac{1 - 2q + q^{d+1}}{1 - q} - q^d = \frac{(1 - 2q)(1 - q^d)}{1 - q} \geq 0.$$

Hence every nonzero difference is at least $2q^k q^d = 2q^N$. $\square$

The theorem gives a clean conditioning baseline for the Vandermonde system in [Theorem 3.2](#). Above $q = 1/2$, no analogous uniform formula is possible: small values of $\{-1, 0, 1\}$ polynomials create near-dependencies long before an exact overlap occurs.

4.3 Submultiplicativity and a signed overlap entropy

Let

$$E_{q,N}(z) := \prod_{j=1}^N 2i \sin(q^j z). \quad (4.6)$$

Its nonzero exponential support has cardinality $r_N(q)$. The geometric scale implies the exact factorization

$$E_{q,N+M}(z) = E_{q,N}(z) E_{q,M}(q^N z). \quad (4.7)$$

Theorem 4.5 (Existence and detection theorem for rank entropy). *For every $0 < q < 1$, the limit*

$$h_{\text{hol}}(q) := \lim_{N \rightarrow \infty} \frac{1}{N} \log r_N(q) = \inf_{N \geq 1} \frac{1}{N} \log r_N(q) \quad (4.8)$$

exists. Equivalently, $\rho_{\text{hol}}(q) := \lim r_N(q)^{1/N}$ exists in $[1, 2]$. Moreover,

$$\boxed{h_{\text{hol}}(q) = \log 2 \iff q \text{ is not a zero of any nonzero polynomial in } \bigcup_{d \geq 0} \mathcal{L}_d.} \quad (4.9)$$

At every exact-overlap parameter, $h_{\text{hol}}(q) < \log 2$.

Proof. If A_N and A_M are the nonzero frequency supports of the two factors on the right of (4.7), then the support of their product is contained in the Minkowski sum $A_N + q^N A_M$. Therefore

$$r_{N+M}(q) \leq r_N(q)r_M(q). \quad (4.10)$$

Fekete's lemma applied to $\log r_N(q)$ proves existence of the limit and the infimum formula.

If there is no restricted-coefficient relation at any level, [Theorem 4.1](#) gives $r_N(q) = 2^N$ for every N , hence $h_{\text{hol}}(q) = \log 2$. Conversely, suppose an exact overlap first occurs at some level N_0 . Then $r_{N_0}(q) < 2^{N_0}$, and the infimum formula gives

$$h_{\text{hol}}(q) \leq \frac{1}{N_0} \log r_{N_0}(q) < \log 2.$$

□

Remark 4.6 (A finite defect has an asymptotic cost). The strict entropy drop is not merely a local curiosity. Once one block of length N_0 has rank below 2^{N_0} , repeated scaled copies of that block force exponential compression at all large scales. Thus a single exact Bernoulli-convolution overlap leaves a permanent asymptotic signature in the minimal ODE orders of the finite sinc products.

Remark 4.7 (Relation to unsigned overlap entropy). Let $d_N(q)$ be the number of *distinct* signed sums, without parity cancellation. The same block argument gives $d_{N+M}(q) \leq d_N(q)d_M(q)$, hence an unsigned support entropy also exists. Always $r_N(q) \leq d_N(q) \leq 2^N$. The signed rank entropy can be strictly smaller because colliding words can cancel. Comparing it with the Garsia entropy of Bernoulli convolutions is one of the main open directions in [Section 12](#).

5 The dyadic case: Thue–Morse spectrum and a closed minimal ODE

5.1 The odd frequency lattice

Set $q = 1/2$ and abbreviate

$$\Phi_N(z) := \prod_{j=1}^N \text{sinc}\left(\frac{z}{2^j}\right), \quad Y_N(z) := z^N \Phi_N(z). \quad (5.1)$$

Write each sign as $\varepsilon_j = 1 - 2b_j$, with $b_j \in \{0, 1\}$, and set $k = \sum_{j=1}^N b_j 2^{N-j}$. Then

$$\sum_{j=1}^N \varepsilon_j 2^{-j} = \frac{2^N - 1 - 2k}{2^N}, \quad \prod_{j=1}^N \varepsilon_j = (-1)^{s_2(k)} =: \tau_k. \quad (5.2)$$

Thus all frequencies are the odd multiples

$$\frac{\pm 1}{2^N}, \frac{\pm 3}{2^N}, \dots, \frac{\pm(2^N - 1)}{2^N}, \quad (5.3)$$

and the coefficient along this ordered lattice is the Thue–Morse sign. Equivalently,

$$Y_N(z) = C_N \sum_{k=0}^{2^N-1} \tau_k \exp\left(\frac{i(2^N - 1 - 2k)z}{2^N}\right), \quad C_N \neq 0. \quad (5.4)$$

The familiar finite product $\sum_{k < 2^N} \tau_k x^k = \prod_{j=0}^{N-1} (1 - x^{2^j})$ is therefore not only a combinatorial identity; it is the complete spectral coefficient vector of the gauged finite Rvachev transform.

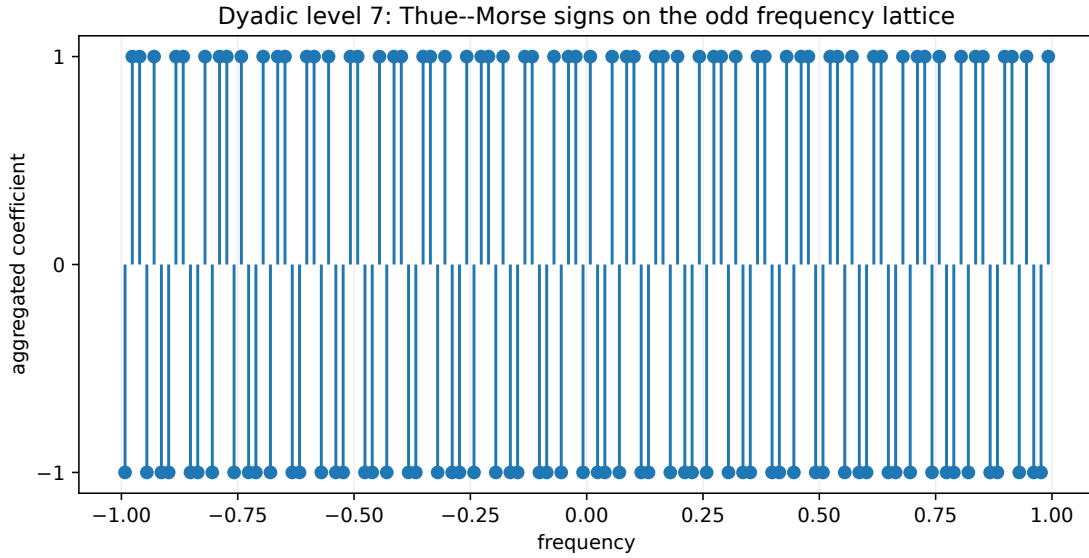


Figure 1: At level $N = 7$, every odd dyadic frequency occurs once, and the coefficient is the Thue–Morse sign $(-1)^{s_2(k)}$.

5.2 The minimal odd-lattice operator

Theorem 5.1 (Closed minimal dyadic ODE). *Let $m = 2^{N-1}$. The gauged product $Y_N = z^N \Phi_N$ satisfies*

$$\mathcal{P}_N(D)Y_N = 0, \quad \mathcal{P}_N(D) := \prod_{\ell=1}^m \left(D^2 + \frac{(2\ell-1)^2}{4^N} \right). \quad (5.5)$$

This equation has order 2^N and is minimal even among homogeneous linear ODEs with rational-function coefficients. Equivalently, Φ_N has the minimal equation

$$z^{-N} \mathcal{P}_N(D) z^N \Phi_N = 0. \quad (5.6)$$

Proof. The roots of the characteristic polynomial in (5.5) are precisely $i(2^N - 1 - 2k)/2^N$, $0 \leq k < 2^N$, the distinct exponents in (5.4). The operator therefore annihilates Y_N . Distinctness gives $r_N(1/2) = 2^N$ by Theorem 4.2; minimality follows from Theorem 3.3. Gauge conjugation yields (5.6) without changing the order. $\square$

The first three operators are

$$\mathcal{P}_1(D) = D^2 + \frac{1}{4}, \quad (5.7)$$

$$\mathcal{P}_2(D) = D^4 + \frac{5}{8}D^2 + \frac{9}{256}, \quad (5.8)$$

$$\mathcal{P}_3(D) = D^8 + \frac{21}{16}D^6 + \frac{987}{2048}D^4 + \frac{3229}{65536}D^2 + \frac{11025}{16777216}. \quad (5.9)$$

The order doubles with every additional box convolution even though the finite products converge locally uniformly to a single entire function.

5.3 A Pochhammer factorization

The odd squares in (5.5) package into a pair of rising factorials.

Corollary 5.2 (Two-sided Pochhammer form). *With $m = 2^{N-1}$ and $(a)_m = a(a+1)\cdots(a+m-1)$,*

$$\mathcal{P}_N(D) = 2^{-(2N-2)m} \left(\frac{1}{2} + i2^{N-1}D \right)_m \left(\frac{1}{2} - i2^{N-1}D \right)_m. \quad (5.10)$$

Proof. Pair the two factors at each ℓ :

$$\begin{aligned} & \left(\ell - \frac{1}{2} + i2^{N-1}D \right) \left(\ell - \frac{1}{2} - i2^{N-1}D \right) \\ &= \left(\ell - \frac{1}{2} \right)^2 + 2^{2N-2}D^2 = 2^{2N-2} \left(D^2 + \frac{(2\ell-1)^2}{4^N} \right). \end{aligned}$$

Multiply over $1 \leq \ell \leq m$. □

This is a literal q -analog bridge only in a broad structural sense: the truncation length $m = 2^{N-1}$ is geometric, while the factorization itself is an ordinary Pochhammer product. The genuinely q -dependent coefficient calculus appears in [Section 6](#).

5.4 All-order Bell–Bernoulli coefficients

Write

$$\mathcal{P}_N(D) = \sum_{r=0}^m A_{N,r} D^{2r}, \quad m = 2^{N-1}. \quad (5.11)$$

The coefficients are elementary symmetric polynomials in the odd squares. Let

$$S_j(m) := \sum_{\ell=1}^m (2\ell-1)^{2j}. \quad (5.12)$$

We use the complete exponential Bell polynomials $\mathcal{B}_k$, normalized by

$$\exp \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{k \geq 0} \mathcal{B}_k(x_1, \dots, x_k) \frac{t^k}{k!}. \quad (5.13)$$

Theorem 5.3 (Bell–Bernoulli coefficient formula). *For $0 \leq r \leq m$, put $k = m - r$. Then*

$$A_{N,r} = 4^{-Nk} \frac{1}{k!} \mathcal{B}_k(S_1(m), -1!S_2(m), \dots, (-1)^{k-1}(k-1)!S_k(m)), \quad (5.14)$$

where every power sum is given explicitly by the Bernoulli-polynomial value

$$S_j(m) = \frac{2^{2j}}{2j+1} \left[B_{2j+1} \left(m + \frac{1}{2} \right) - B_{2j+1} \left(\frac{1}{2} \right) \right]. \quad (5.15)$$

Thus every coefficient of the minimal dyadic ODE is available to arbitrary order by a finite Bell-polynomial formula.

Proof. Expanding the product in (5.5) gives

$$A_{N,r} = 4^{-N(m-r)} e_{m-r}(1^2, 3^2, \dots, (2m-1)^2),$$

where e_k denotes the elementary symmetric polynomial. Its generating function is

$$\sum_{k=0}^m e_k t^k = \prod_{\ell=1}^m (1 + (2\ell-1)^2 t)$$

$$= \exp \left(\sum_{j \geq 1} \frac{(-1)^{j-1}}{j} S_j(m) t^j \right).$$

Comparing with (5.13) proves (5.14). Finally,

$$S_j(m) = 2^{2j} \sum_{\ell=1}^m \left(\ell - \frac{1}{2} \right)^{2j},$$

and the standard Bernoulli-polynomial summation formula $\sum_{\ell=1}^m (\ell + a)^p = [B_{p+1}(m + a + 1) - B_{p+1}(a + 1)]/(p + 1)$ gives (5.15) after the index is shifted. $\square$

Remark 5.4 (A closed coefficient pipeline). The theorem gives an exact symbolic pipeline

Bernoulli polynomials $\longrightarrow$ odd power sums $\longrightarrow$ Bell polynomials $\longrightarrow$ minimal ODE coefficients.

It is independent of the numerical size 2^N of the frequency list. This provides a new role for the Bell–Bernoulli machinery already prominent in the moment and endpoint parts of the repository.

5.5 Discriminant and Barnes- G asymptotics

Let $n = 2^N$ and let

$$\chi_N(\lambda) := \prod_{k=0}^{n-1} \left(\lambda - \frac{i(n-1-2k)}{n} \right) \quad (5.16)$$

be the monic characteristic polynomial of the minimal constant-coefficient operator. Its roots are an arithmetic progression of step $2i/n$.

Theorem 5.5 (Exact dyadic discriminant). *The absolute discriminant of χ_N is*

$$|\text{Disc } \chi_N| = \left(\frac{2}{n} \right)^{n(n-1)} \prod_{j=1}^{n-1} (j!)^2 = \left(\frac{2}{n} \right)^{n(n-1)} G(n+1)^2, \quad (5.17)$$

where G is the Barnes G -function. As $n = 2^N \rightarrow \infty$,

$$\log |\text{Disc } \chi_N| = n^2 \left(\log 2 - \frac{3}{2} \right) + n \log(\pi n) - \frac{1}{6} \log n + 2\zeta'(-1) + O(n^{-2}). \quad (5.18)$$

Proof. For an arithmetic progression $a + hk$, $0 \leq k < n$,

$$\prod_{0 \leq j < k < n} |h(k-j)|^2 = |h|^{n(n-1)} \prod_{k=1}^{n-1} (k!)^2.$$

Here $|h| = 2/n$, which proves the first equality. The defining integer identity $G(n+1) = 1!2! \cdots (n-1)!$ proves the second. Substitution of the standard expansion

$$\log G(n+1) = \frac{n^2}{2} \log n - \frac{3n^2}{4} + \frac{n}{2} \log(2\pi) - \frac{1}{12} \log n + \zeta'(-1) + O(n^{-2})$$

into (5.17) yields (5.18) after cancellation of the $n^2 \log n$ terms. $\square$

The negative quadratic coefficient $\log 2 - 3/2$ shows that the discriminant becomes exponentially small in the square of the differential rank. Exact rank is therefore not the whole numerical story: even the perfectly separated dyadic lattice leads to increasingly ill-conditioned coefficient recovery in monomial bases.

6 General- q operator coefficients from q -integers and Bell polynomials

Even when collisions occur, there is a canonical order- 2^N operator obtained by retaining the entire sign-word multiset. Define

$$\mathcal{P}_{q,N}(\lambda) := \prod_{\varepsilon \in \{\pm 1\}^N} \left(\lambda - i \sum_{j=1}^N \varepsilon_j q^j \right). \quad (6.1)$$

It annihilates $Y_{q,N}$. It is minimal precisely at full-rank parameters; at an overlap parameter the minimal polynomial is obtained by retaining each nonzero aggregated frequency once.

Proposition 6.1 (Scale recursion). *The universal characteristic polynomials satisfy*

$$\mathcal{P}_{q,N+1}(\lambda) = \mathcal{P}_{q,N}(\lambda - iq^{N+1}) \mathcal{P}_{q,N}(\lambda + iq^{N+1}). \quad (6.2)$$

Proof. Every level- $(N+1)$ frequency is a level- N frequency plus or minus q^{N+1} . Split the product according to the last sign. $\square$

The coefficients can also be built from cumulants. Let B_{2r} denote the Bernoulli numbers with $B_2 = 1/6$, and write

$$[N]_x := \frac{1 - x^N}{1 - x}. \quad (6.3)$$

Theorem 6.2 (q -integer cumulants of the frequency multiset). *For the Rademacher sum $W_{q,N}$ in (2.7), all odd cumulants vanish and*

$$\kappa_{2r}(W_{q,N}) = \frac{2^{2r}(2^{2r} - 1)B_{2r}}{2r} q^{2r} [N]_{q^{2r}}. \quad (6.4)$$

Consequently, all raw moments of the frequency multiset are

$$\mathbb{E}W_{q,N}^k = \mathcal{B}_k(0, \kappa_2, 0, \kappa_4, \dots, \kappa_k), \quad (6.5)$$

and the power sums of the roots of (6.1) are

$$p_k := \sum_{\varepsilon} (i\omega(\varepsilon))^k = 2^N i^k \mathbb{E}W_{q,N}^k. \quad (6.6)$$

Newton's identities, equivalently a second Bell-polynomial transform, recover every coefficient of $\mathcal{P}_{q,N}$ from these p_k .

Proof. Independence gives

$$\log \mathbb{E}e^{tW_{q,N}} = \sum_{j=1}^N \log \cosh(tq^j). \quad (6.7)$$

The Bernoulli expansion

$$\log \cosh x = \sum_{r \geq 1} \frac{2^{2r}(2^{2r} - 1)B_{2r}}{2r(2r)!} x^{2r} \quad (6.8)$$

shows that the $(2r)$ th cumulant is the displayed Bernoulli factor times $\sum_{j=1}^N q^{2rj} = q^{2r} [N]_{q^{2r}}$. The moment–cumulant formula is exactly (6.5). Summing the k th powers over all 2^N equally weighted sign words gives (6.6); Newton's identities are standard. $\square$

Corollary 6.3 (Full-rank coefficient formula). *If q is not a root of a nonzero polynomial in $\mathcal{L}_{N-1}$, then (6.1) is the minimal characteristic polynomial of $Y_{q,N}$. Hence Theorem 6.2 is an all-order formula for the coefficients of the minimal ODE in terms of Bernoulli numbers, complete Bell polynomials, and the finite q -integers $[N]_{q^{2r}}$.*

Proof. The hypothesis gives full rank by [Theorem 4.1](#); therefore all 2^N roots in (6.1) are distinct and required by [Theorem 3.3](#). $\square$

This combines four strands usually treated separately: geometric scaling, Rademacher convolutions, Bernoulli-number cumulants, and reconstruction of the differential operator by Bell polynomials.

7 The inverse-golden-ratio overlap: the first rank defect

Let

$$q = \varphi^{-1} = \frac{\sqrt{5}-1}{2}, \quad q^2 + q = 1. \quad (7.1)$$

Then $q - q^2 - q^3 = q(1 - q - q^2) = 0$. The corresponding collision appears at the first possible nontrivial level.

Proposition 7.1 (Exact level-three golden defect). *At $q = \varphi^{-1}$,*

$$r_1(q) = 2, \quad r_2(q) = 4, \quad \boxed{r_3(q) = 6}. \quad (7.2)$$

The two sign words $(+, -, -)$ and $(-, +, +)$ both have frequency zero and have opposite aggregate signs, so the central atom cancels completely. The six surviving frequencies are

$$\pm 2q, \quad \pm 2q^2, \quad \pm 2q^3. \quad (7.3)$$

Proof. The displayed polynomial relation gives $q - q^2 - q^3 = 0$ and $-q + q^2 + q^3 = 0$. The products of the signs are respectively $+1$ and -1 , hence cancellation at zero. Direct use of $q^2 = 1 - q$ reduces the remaining six sign sums to the six nonzero numbers in (7.3); they are pairwise distinct. Apply [Theorem 3.3](#). $\square$

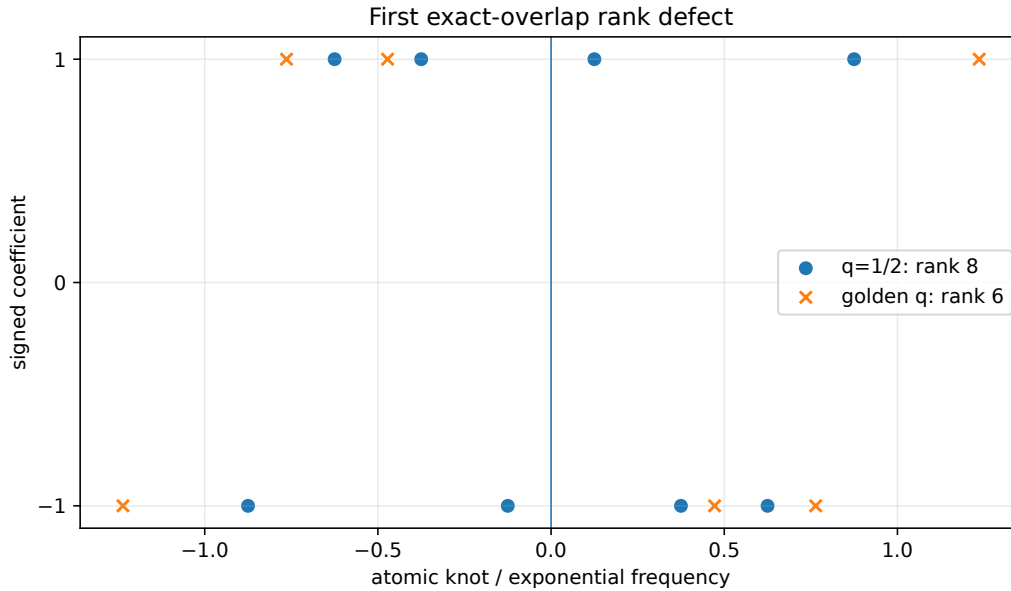


Figure 2: The level-three dyadic product has eight signed atoms. At the golden parameter, the two central words collide at zero with opposite parity and disappear, leaving differential rank six.

By [Theorem 4.5](#), even this single finite identity forces

$$h_{\text{hol}}(\varphi^{-1}) \leq \frac{1}{3} \log 6 < \log 2. \quad (7.4)$$

Exact quadratic-field computations reveal a much more rigid pattern.

Conjecture 7.2 (Golden signed-rank recurrence). *Let $r_N = r_N(\varphi^{-1})$ and set $r_0 = 1$. Then*

$$\boxed{r_N = 2r_{N-1} - 2r_{N-2} + 2r_{N-3} \quad (N \geq 3),} \quad (7.5)$$

with $r_0, r_1, r_2 = 1, 2, 4$. Equivalently,

$$\sum_{N \geq 0} r_N x^N = \frac{1 + 2x^2}{1 - 2x + 2x^2 - 2x^3}. \quad (7.6)$$

If true, the holonomic growth base is the real root

$$\alpha = 1.543689012692 \dots \quad \text{of} \quad \alpha^3 - 2\alpha^2 + 2\alpha - 2 = 0, \quad (7.7)$$

so $h_{\text{hol}}(\varphi^{-1}) = \log \alpha$.

Exact computational evidence. Arithmetic was performed in $\mathbb{Z}[q]/(q^2 + q - 1)$, representing every frequency as $a + bq$ with integers a, b . Through level 25 there was no failure of (7.5); the ranks were

$$\begin{aligned} &1, 2, 4, 6, 8, 12, 20, 32, 48, 72, 112, 176, 272, 416, 640, 992, \\ &1536, 2368, 3648, 5632, 8704, 13440, 20736, 32000, 49408, 76288. \end{aligned} \quad (7.8)$$

Every surviving aggregate coefficient had absolute value one. These are exact integer statements, but they do not supply an induction invariant; hence the recurrence remains a conjecture.

The unsigned numbers of distinct sums through the same range agree with $F_{N+3} - 1$, where F_n is Fibonacci. This suggests a finite-state normal form for binary words under the local relation $100 \leftrightarrow 011$. A proof of the signed recurrence likely requires a refinement that records the parity of each equivalence class, not merely its cardinality.

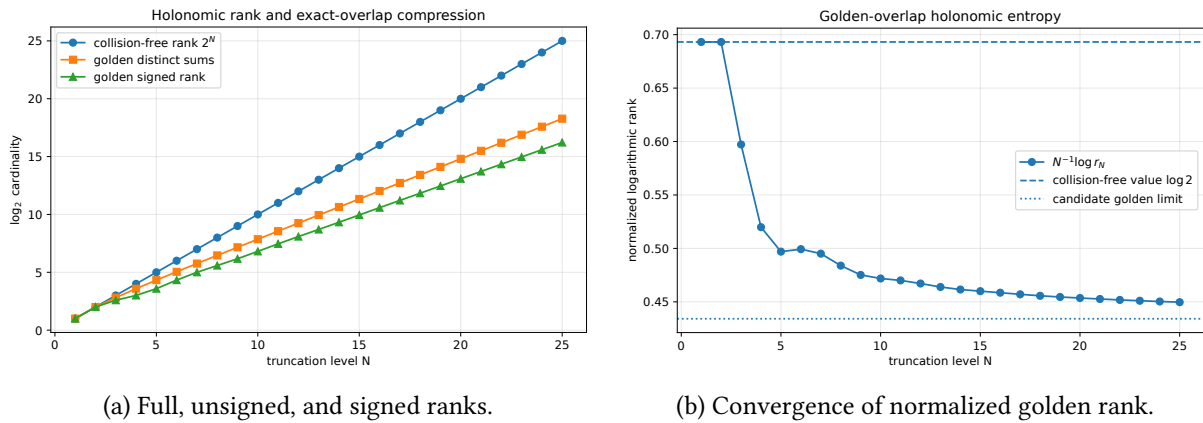


Figure 3: Exact overlap compresses both the distinct-sum set and, more strongly, the signed holonomic support.

8 The infinite transform: first-order dilation, infinite differential rank

8.1 Integer-base sinc products and their divisor

For an integer $b \geq 2$, define

$$\Phi_b(z) := \prod_{j=1}^{\infty} \text{sinc}\left(\frac{z}{b^j}\right). \quad (8.1)$$

The product converges locally uniformly because $\text{sinc}(z/b^j) = 1 + O_K(b^{-2j})$ on every compact set $K \subset \mathbb{C}$. It obeys the first-order dilation equation

$$\boxed{\Phi_b(bz) = \text{sinc}(z)\Phi_b(z)}. \quad (8.2)$$

This is an extremely low-complexity equation in the dilation operator $f(z) \mapsto f(bz)$. Differential complexity behaves in the opposite way.

The zero divisor is elementary and already belongs to the repository's spectral layer; we record it to make the new deduction self-contained.

Proposition 8.1 (Integer-base zero multiplicities). *The nonzero zeros of Φ_b are the points*

$$z = \pi b m, \quad m \in \mathbb{Z} \setminus \{0\}, \quad (8.3)$$

and

$$\text{ord}_{z=\pi b m} \Phi_b = 1 + \nu_b(m), \quad (8.4)$$

where $\nu_b(m) = \max\{k : b^k \mid m\}$.

Proof. The j th factor vanishes at $z = \pi n b^j$, $n \neq 0$, and all its nonzero zeros are simple. At $z = \pi b m$, the factor indexed by j vanishes exactly when b^{j-1} divides m , namely for $1 \leq j \leq 1 + \nu_b(m)$. No other factor vanishes there. $\square$

8.2 Non-D-finiteness from unbounded multiplicity

Theorem 8.2 (The infinite sinc product is not D-finite). *For every integer $b \geq 2$, the entire function Φ_b is not D-finite over $\mathbb{C}(z)$. In particular, it is transcendental over $\mathbb{C}(z)$.*

Proof. Suppose Φ_b satisfied an order- r equation (2.8), with leading polynomial $p_r \neq 0$. The zeros $z_k = \pi b^{k+1}$ have multiplicities $k+1$ by Theorem 8.1. For infinitely many $k \geq r$, $p_r(z_k) \neq 0$ because a nonzero polynomial has only finitely many zeros. At such an ordinary point, Φ_b has zero multiplicity at least r , contradicting Theorem 2.3. Algebraic functions are D-finite, so non-D-finiteness implies transcendence over $\mathbb{C}(z)$. $\square$

Corollary 8.3 (A locally uniform limit leaves the D-finite category). *For $b = 2$, every truncation Φ_N is D-finite of exact rank 2^N , and $\Phi_N \rightarrow \Phi_2$ locally uniformly on $\mathbb{C}$, but the limit is not D-finite. Thus neither D-finiteness nor bounded differential rank is preserved under this natural locally uniform approximation.*

Proof. Finite-rank D-finiteness is Theorem 5.1; local uniform convergence follows from the product estimate above; non-D-finiteness is Theorem 8.2. $\square$

Remark 8.4 (Difference order versus differential order). Equation (8.2) has order one in a dilation Ore algebra, while no finite-order linear differential operator over $\mathbb{C}(z)$ annihilates the same function. The Fabius–Rvachev transform is therefore a particularly explicit separator between dilation-holonomic and differential-holonomic behavior.

8.3 Escape from every finite dyadic trigonometric field

For $M \geq 0$, let

$$\mathcal{T}_M := \mathbb{C}\left(z, e^{iz/2^M}\right). \quad (8.5)$$

These fields form an increasing tower and are stable under D. The finite logarithmic derivatives belong to it:

$$g_N(z) := \frac{\Phi'_N(z)}{\Phi_N(z)} = \sum_{j=1}^N \left(2^{-j} \cot\left(\frac{z}{2^j}\right) - \frac{1}{z} \right) \in \mathcal{T}_N. \quad (8.6)$$

Let $g = \Phi'_2/\Phi_2$. Its pole at $2\pi m$ is simple and has residue $1 + \nu_2(m)$.

We first isolate a rational-function lemma.

Lemma 8.5 (Residues along a periodic rational curve). *Fix $M \geq 0$ and let $R(z, w) \in \mathbb{C}(z, w)$. Suppose $R(z, e^{iz/2^M})$ has a simple pole at $z_n = 2\pi 2^M n$ for all sufficiently large positive integers n . Then its residue at z_n agrees, outside a finite exceptional set, with a rational function of n .*

Proof. Expand R as a Laurent series at the divisor $w = 1$ over the coefficient field $\mathbb{C}(z)$:

$$R(z, w) = \sum_{k=-s}^{\infty} A_k(z)(w-1)^k, \quad A_{-s} \neq 0, \quad (8.7)$$

where the principal part is finite and every $A_k(z)$ is rational. A rational factor depending only on z can create poles at only finitely many z_n ; therefore $s \geq 1$. Since

$$e^{iz/2^M} - 1 = \frac{i}{2^M}(z - z_n) + O((z - z_n)^2) \quad (8.8)$$

near z_n , the generic pole order after substitution is s . The coefficient $A_{-s}(z_n)$ can vanish for only finitely many n unless it is identically zero. The assumed simple poles therefore force $s = 1$. Their residues are

$$\frac{2^M}{i} A_{-1}(2\pi 2^M n), \quad (8.9)$$

which is a rational function of n . $\square$

Theorem 8.6 (Trigonometric-field escape). *For every $M \geq 0$,*

$$\boxed{\frac{\Phi'_2}{\Phi_2} \notin \mathcal{T}_M}. \quad (8.10)$$

Consequently,

$$\Phi_2 \notin \bigcup_{M \geq 0} \mathcal{T}_M. \quad (8.11)$$

Thus the infinite logarithmic derivative is not a rational expression in z and trigonometric functions from any fixed finite dyadic frequency level, even though every finite truncation has such an expression.

Proof. Suppose $g = \Phi'_2/\Phi_2$ belonged to $\mathcal{T}_M$. At $z_n = 2\pi 2^M n$ its residue is

$$1 + \nu_2(2^M n) = M + 1 + \nu_2(n). \quad (8.12)$$

By Theorem 8.5, this would agree eventually with a rational function $S(n) \in \mathbb{C}(n)$. On all sufficiently large odd integers n , the right-hand side of (8.12) equals $M + 1$. Hence the rational function $S(n) - (M + 1)$ has infinitely many zeros and is identically zero. But at $n = 2^k$ the required residue is $M + 1 + k$, a contradiction.

If $\Phi_2 \in \mathcal{T}_M$, then stability under differentiation and division would imply $\Phi'_2/\Phi_2 \in \mathcal{T}_M$, already excluded. $\square$

Remark 8.7. The proof uses not merely the existence of infinitely many zeros but the non-rational arithmetic variation of their multiplicities along the lattice. The ruler function $1 + \nu_2(n)$ is therefore an obstruction to finite-level trigonometric closed form.

9 Moment, cumulant, and reciprocal sequences are not P-recursive

9.1 The moment-generating function

For the dyadic random variable X in (2.3), define

$$M(t) := \mathbb{E}e^{tX} = \prod_{j=1}^{\infty} \frac{\sinh(t/2^j)}{t/2^j} = \Phi_2(-it). \quad (9.1)$$

Let

$$M(t) = \sum_{n \geq 0} \mu_n \frac{t^n}{n!}, \quad K(t) := \log M(t) = \sum_{n \geq 1} \kappa_n \frac{t^n}{n!}, \quad \frac{1}{M(t)} = \sum_{n \geq 0} \rho_n \frac{t^n}{n!}. \quad (9.2)$$

Here μ_n are moments, κ_n are cumulants, and ρ_n are the signed reciprocal-moment coefficients. Symmetry makes all odd terms vanish.

The cumulants have a familiar exact Bernoulli formula, included both for reference and to expose the Bell-polynomial consequences.

Proposition 9.1 (Rvachev cumulants). *For $r \geq 1$,*

$$\boxed{\kappa_{2r} = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0.} \quad (9.3)$$

Furthermore,

$$\mu_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad \rho_n = \mathcal{B}_n(-\kappa_1, \dots, -\kappa_n). \quad (9.4)$$

Proof. The standard expansion

$$\log \frac{\sinh x}{x} = \sum_{r \geq 1} \frac{2^{2r} B_{2r}}{2r(2r)!} x^{2r} \quad (9.5)$$

gives

$$\begin{aligned} K(t) &= \sum_{j \geq 1} \log \frac{\sinh(t/2^j)}{t/2^j} \\ &= \sum_{r \geq 1} \frac{2^{2r} B_{2r}}{2r(2r)!} \left(\sum_{j \geq 1} 2^{-2rj} \right) t^{2r} \\ &= \sum_{r \geq 1} \frac{2^{2r} B_{2r}}{2r(2r)!(2^{2r} - 1)} t^{2r}. \end{aligned}$$

Comparison with (9.2) proves (9.3). Since $M = \exp K$ and $M^{-1} = \exp(-K)$, the two Bell formulae follow from (5.13). $\square$

9.2 Three non-P-recursive rational sequences

Theorem 9.2 (Moment and Bell-sequence obstruction). *None of the three sequences*

$$(\mu_n)_{n \geq 0}, \quad (\kappa_n)_{n \geq 0}, \quad (\rho_n)_{n \geq 0} \quad (9.6)$$

is P-recursive. Their even subsequences (μ_{2n}) , (κ_{2n}) , and (ρ_{2n}) are likewise not P-recursive.

Proof. The MGF $M(t) = \Phi_2(-it)$ is not D-finite by Theorem 8.2. If (μ_n) were P-recursive, then $(\mu_n/n!)$ would also be P-recursive: substitute $\mu_{n+j} = (n+j)! [t^{n+j}]M$ into a recurrence and clear shifted factorials. Stanley's equivalence would make M D-finite, a contradiction.

The function M has infinitely many zeros, inherited from Φ_2 . Therefore $1/M$ has infinitely many poles and is not D-finite by Theorem 2.4. This excludes P-recursive-ness of (ρ_n) by the same factorial argument.

If K were D-finite, then $K' = M'/M$ would be D-finite. But M'/M has a simple pole at every zero of M , with residue equal to its multiplicity, hence infinitely many finite-plane poles. This contradicts Theorem 2.4. Thus (κ_n) is not P-recursive.

Finally suppose, for example, that (μ_{2n}) were P-recursive. Then $c_n = \mu_{2n}/(2n)!$ would be P-recursive after clearing the hypergeometric factor $(2n)!$ in any recurrence. The ordinary series $C(z) = \sum c_n z^n$ would be D-finite, and D-finiteness is preserved under the algebraic substitution $z = t^2$. But $C(t^2) = M(t)$, a contradiction. The same argument applies to the other two even subsequences. $\square$

Corollary 9.3 (A non-P-recursive Bernoulli-number specialization). *The explicit rational sequence*

$$\left(\frac{2^{2n} B_{2n}}{2n(2^{2n} - 1)} \right)_{n \geq 1} \quad (9.7)$$

is not P-recursive.

Proof. It is the even cumulant sequence in (9.3); apply Theorem 9.2. $\square$

Remark 9.4 (Why explicit does not mean holonomic). The cumulants have a one-line Bernoulli formula, and the moments and reciprocal coefficients have finite Bell-polynomial formulae at each order. Nevertheless, no fixed polynomial-coefficient recurrence controls any of the resulting infinite sequences. The obstruction is global analytic structure: infinitely many singularities for K' and $1/M$, and unbounded zero multiplicity for M .

9.3 The functions F and up themselves

Fabius's original example is nowhere analytic on its active interval [7]; the repository develops stronger variants and iterated consequences. This immediately has a holonomic interpretation.

Corollary 9.5 (The Fabius and up functions are not D-finite). *No nonzero polynomial-coefficient linear ODE can hold for F on an interval containing a nontrivial subinterval of $(0, 1)$. The same is true for up on an interval meeting $(-1, 1)$ nontrivially.*

Proof. A solution of a polynomial-coefficient linear ODE is real analytic at every ordinary point; only the finitely many roots of the leading coefficient can be singular. The function F is nowhere analytic on $(0, 1)$, so such an ODE is impossible. Relation (2.2) transfers the same obstruction to either active half of up . $\square$

10 Deep dyadic Fabius samples and P-recursiveness

The previous section used the Fourier transform. A different obstruction comes directly from the endpoint self-differential equation and applies to the exact sample sequence most prominent in the arithmetic theory.

For fixed $c > 0$, define, after any harmless finite initial prefix,

$$a_n(c) := F(c2^{-n}). \quad (10.1)$$

For all sufficiently large n , the argument lies in $(0, 1/2]$. Integrating (2.1) from 0 to x and using $F(0) = 0$ gives the exact Volterra identity

$$F(x) = \int_0^{2x} F(u) \, du, \quad 0 \leq x \leq \frac{1}{2}. \quad (10.2)$$

Monotonicity now forces a rapidly contracting ratio.

Lemma 10.1 (Supergeometric one-step bound). *For every fixed $c > 0$ and all sufficiently large n ,*

$$0 < a_{n+1}(c) \leq c2^{-n} a_n(c). \quad (10.3)$$

More generally, for every fixed $j \geq 1$,

$$\frac{a_{n+j}(c)}{a_n(c)} \leq c^j 2^{-jn-j(j-1)/2}. \quad (10.4)$$

Proof. Apply (10.2) with $x = c2^{-n-1}$:

$$a_{n+1}(c) = \int_0^{c2^{-n}} F(u) \, du.$$

Since F is increasing and positive on $(0, 1)$, the integral is positive and at most $c2^{-n}F(c2^{-n})$. Iterating this estimate proves (10.4). $\square$

Theorem 10.2 (No P-recursion for deep dyadic samples). *For every $c > 0$, the sequence $(F(c2^{-n}))_{n \geq 0}$ is not P-recursive. In particular, the exact rational sequence*

$$\boxed{(F(2^{-n}))_{n \geq 0}} \quad (10.5)$$

is not P-recursive.

Proof. A finite change of initial terms does not affect P-recursiveness, so assume all arguments are in $(0, 1/2]$. Suppose

$$\sum_{j=0}^r p_j(n) a_{n+j}(c) = 0 \quad (10.6)$$

for all sufficiently large n , with polynomial coefficients not all zero. Let j_0 be the least index for which p_{j_0} is nonzero. Divide by the positive number $a_{n+j_0}(c)$. By (10.4), for $j > j_0$,

$$\frac{a_{n+j}(c)}{a_{n+j_0}(c)} = O\left(2^{-(j-j_0)n}\right). \quad (10.7)$$

Therefore (10.6) implies

$$p_{j_0}(n) = - \sum_{j=j_0+1}^r p_j(n) \frac{a_{n+j}(c)}{a_{n+j_0}(c)} \longrightarrow 0. \quad (10.8)$$

The right-hand side tends to zero because a polynomial times an exponentially decaying sequence tends to zero. A nonzero polynomial cannot tend to zero along the positive integers. This contradiction proves the theorem.

Arias de Reyna proved that F takes rational values at dyadic arguments and established strong denominator and integrality properties [2, 3]. Hence the specialization $c = 1$ is an exact rational, not merely numerical, example. $\square$

10.1 All fixed derivative orders

Repeated differentiation of (2.1) gives a triangular scaling law near the endpoint.

Lemma 10.3 (Endpoint derivative scaling). *For every integer $m \geq 0$ and $0 \leq x \leq 2^{-m}$,*

$$F^{(m)}(x) = 2^{m(m+1)/2} F(2^m x). \quad (10.9)$$

Proof. The assertion is trivial for $m = 0$ and is (2.1) for $m = 1$. If it holds for m , differentiate on $0 \leq x \leq 2^{-(m+1)}$:

$$\begin{aligned} F^{(m+1)}(x) &= 2^{m(m+1)/2} 2^m F'(2^m x) \\ &= 2^{m(m+1)/2+m+1} F(2^{m+1} x) \\ &= 2^{(m+1)(m+2)/2} F(2^{m+1} x). \end{aligned}$$

$\square$

Corollary 10.4 (Derivative sample obstruction). *For every fixed $m \geq 0$ and $c > 0$, the sequence*

$$(F^{(m)}(c2^{-n}))_{n \geq 0} \quad (10.10)$$

is not P-recursive.

Proof. For sufficiently large n , (10.9) gives

$$F^{(m)}(c2^{-n}) = 2^{m(m+1)/2} F(c2^{-(n-m)}).$$

This is a nonzero constant multiple and finite shift of the sequence in [Theorem 10.2](#). $\square$

10.2 An entire generating function of order zero

Define the ordinary generating function

$$A_c(z) := \sum_{n=0}^{\infty} F(c2^{-n}) z^n, \quad (10.11)$$

where any clamped or omitted finite prefix is immaterial.

Theorem 10.5 (Order-zero sample generating function). *For every $c > 0$, A_c is a nonpolynomial entire function of order zero. More precisely, with $M_c(r) = \max_{|z|=r} |A_c(z)|$,*

$$\log M_c(r) \leq \frac{(\log r)^2}{2 \log 2} + O(\log r) \quad (r \rightarrow \infty). \quad (10.12)$$

It is not D-finite.

Proof. Iteration of (10.3) yields constants $C_0, C_1 > 0$ such that

$$0 < a_n(c) \leq C_0 C_1^n 2^{-n(n-1)/2}. \quad (10.13)$$

The ratio test gives infinite radius of convergence. For $r \geq 1$,

$$M_c(r) \leq C_0 \sum_{n \geq 0} \exp \left[-\frac{\log 2}{2} n^2 + n \left(\log(C_1 r) + \frac{\log 2}{2} \right) \right].$$

Completing the square bounds this Gaussian sum by a polynomial factor times

$$\exp \left(\frac{(\log(C_1 r) + \frac{1}{2} \log 2)^2}{2 \log 2} \right),$$

which proves (10.12). Consequently

$$\limsup_{r \rightarrow \infty} \frac{\log \log M_c(r)}{\log r} = 0,$$

so the order is zero. Every coefficient is eventually positive, hence the function is not a polynomial. Finally, if A_c were D-finite, its coefficient sequence would be P-recursive, contrary to [Theorem 10.2](#). $\square$

Remark 10.6 (Endpoint asymptotics enter at the next order). The leading log-square bound in (10.12) is only an upper estimate. The repository's corrected endpoint transseries for $F(x)$ as $x \downarrow 0$, with Lambert- W carrier and a nonconstant dyadic periodic factor, should permit a saddle-point expansion of $A_c(r)$ with logarithmic periodicity. This is formulated as a future problem in [Theorem 12.6](#).

11 Numerical experiments and reproducibility

All experiments are contained in the commented script `frontier_experiments.py`. The rank computations use exact arithmetic: $\mathbb{Q}$ for rational q , and integer pairs $a + bq$ modulo $q^2 + q - 1$ for the golden parameter. Floating point is used only to order already deduplicated algebraic numbers for plotting, and to display high-precision residuals.

11.1 Rank table

Table 1: Exact ranks and unsigned distinct-sum counts at selected levels.

N	$r_N(1/2)$	$r_N(2/3)$	$r_N(\varphi^{-1})$	$d_N(\varphi^{-1})$
1	2	2	2	2
2	4	4	4	4
3	8	8	6	7
4	16	16	8	12
5	32	32	12	20
6	64	64	20	33
8	256	256	48	88
10	1024	1024	112	232
12	4096	4096	272	609
16	65536	65536	1536	4180
20	1048576	1048576	8704	28656
25	33554432	33554432	76288	317810

The rational parameter $2/3$ stays at full exact rank, as proved by the rational-root argument in [Theorem 4.2](#). Nevertheless its frequency separation shrinks faster than in the dyadic case, illustrating that exact rank and numerical conditioning are distinct invariants.

11.2 Operator residual and discriminant checks

The script constructs $\mathcal{P}_N$ with exact SymPy rationals and evaluates $\mathcal{P}_N(D)Y_N$ by differentiating the Thue–Morse exponential sum term by term at 120 decimal digits. For $1 \leq N \leq 5$ and the test points $z = 0.37, 1.1, 2.7$, the largest relative residual was below

$$8.4 \times 10^{-122}. \quad (11.1)$$

This is a consistency check, not part of the proof of minimality.

The Barnes- G discriminant formula was evaluated through $N = 12$ using [\(5.17\)](#); the difference from the asymptotic [\(5.18\)](#) decays at the expected inverse-square rate. Complete tables are included under `data/`.

11.3 Reproduction command

From the unpacked archive, run

```
python frontier_experiments.py --output-dir data
latexmk -pdf -interaction=nonstopmode fabius_holonomic_frontiers.tex
```

The script records the tested package versions in `NUMERICAL_README.md`; `requirements.txt` gives the minimal Python dependencies.

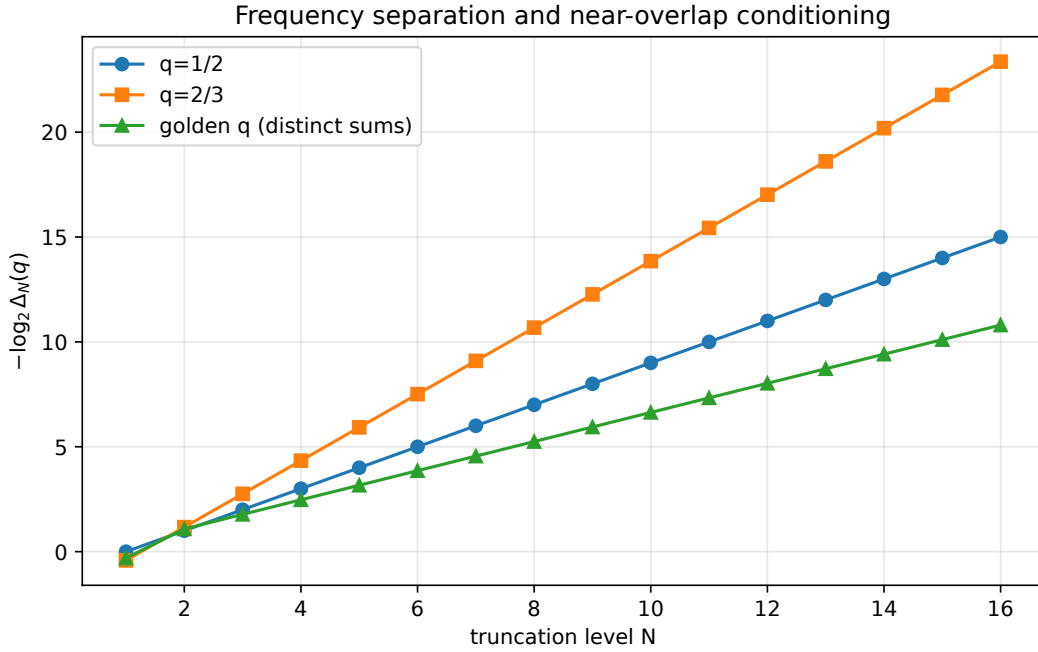


Figure 4: Minimum spacing of distinct signed sums. For $q = 1/2$, the exact law is $\Delta_N = 2^{1-N}$. The $q = 2/3$ and golden curves were computed in exact rational or quadratic-field arithmetic before numerical display.

12 Conjectures and promising research directions

12.1 Proving the golden finite-state recurrence

Problem 12.1 (A parity-sensitive normalization automaton). *Construct a finite-state transducer for sign words modulo $q^2 + q - 1 = 0$ that tracks both the represented frequency and the parity weight $\prod \varepsilon_j$. Use it to prove or refute Theorem 7.2, and determine whether every surviving aggregate coefficient is always ± 1 .*

The unsigned relation $100 \leftrightarrow 011$ suggests a canonical-word language, but signed cancellation requires more information than a normal form for values. A successful automaton should yield a rational generating function for r_N , a finite matrix whose Perron root is the rank growth base, and perhaps a complete classification of local knot cancellations in the finite splines.

12.2 Differential transcendence

Conjecture 12.2 (Differential transcendence of the Rvachev transform). *For every integer $b \geq 2$, the function*

$$\Phi_b(z) = \prod_{j \geq 1} \text{sinc}(z/b^j) \quad (12.1)$$

is differentially transcendental over $\mathbb{C}(z)$; that is, it satisfies no nonzero algebraic differential equation with coefficients in $\mathbb{C}(z)$. The same should hold for its moment-generating specialization $\Phi_b(-it)$.

Theorem 8.2 excludes only *linear* differential equations. The first-order dilation equation (8.2) makes the conjecture a natural target for parameterized difference Galois theory. Existing criteria cover broad classes of shift, q -dilation, and Mahler equations [4, 1], but the coefficient $\text{sinc } z$ naturally lives in a trigonometric extension rather than $\mathbb{C}(z)$. The residue obstruction in Theorem 8.6 is evidence that no finite trigonometric extension trivializes the equation.

A concrete intermediate target is to classify all differential-algebraic solutions of

$$y(2z) = \text{sinc}(z)y(z) \quad (12.2)$$

over the differential-difference field generated by z and all $e^{iz/2^M}$. The divisor orbit under $z \mapsto 2z$ should enter the parameterized Galois group explicitly.

12.3 Rank entropy versus Bernoulli-convolution entropy

Conjecture 12.3 (Signed and unsigned overlap exponents). *Let*

$$h_{\text{supp}}(q) = \lim_{N \rightarrow \infty} \frac{1}{N} \log d_N(q), \quad h_{\text{hol}}(q) = \lim_{N \rightarrow \infty} \frac{1}{N} \log r_N(q). \quad (12.3)$$

For algebraic exact-overlap parameters, both exponents should be computable by a finite algebraic automaton. The signed exponent should equal the logarithm of the spectral radius of a parity-refined overlap matrix, and should admit inequalities in terms of the Garsia entropy of the corresponding Bernoulli convolution.

The theorem $h_{\text{hol}}(q) < \log 2$ if and only if an exact overlap occurs is stronger than a generic upper bound but weaker than a quantitative entropy formula. Hochman’s inverse theorem identifies superexponentially close cylinders as the mechanism behind dimension drop and proves exact-overlap consequences for algebraic parameters [8]. The new rank entropy is a signed algebraic statistic that may provide a tractable finite-state shadow of that theory.

12.4 Near-overlap conditioning and restricted-coefficient polynomials

The exact identity (4.4) turns the smallest frequency gap into a restricted-coefficient polynomial problem. This suggests three quantitative projects.

1. Bound the condition number of the derivative-to-frequency Vandermonde matrix in terms of $\Delta_N(q)$ and the maximum frequency.
2. For a fixed algebraic q with no exact overlap, combine root-separation bounds with the restricted-coefficient literature [6, 5] to obtain explicit lower bounds for $\Delta_N(q)$.
3. Study a numerical “effective rank” obtained by merging frequencies below a precision threshold. Its phase changes should track near roots of $\{-1, 0, 1\}$ polynomials even when the exact rank remains 2^N .

12.5 A mixed dilation–differential Ore algebra

The finite operators $\mathcal{P}_N(D)$ grow exponentially in order, while the infinite transform has the first-order dilation equation (8.2). A computationally useful symbolic theory should therefore not insist on pure differential holonomicity.

Problem 12.4 (Minimal mixed operator). *In the Ore algebra generated by multiplication by meromorphic functions, D , and the dilation $\sigma_b f(z) = f(bz)$, determine a canonical minimal annihilating ideal for Φ_b . Develop closure and creative-telescoping rules that retain the first-order dilation relation instead of expanding to the exponentially large finite differential operators.*

The finite rank formulae provide test cases: truncation at level N should produce an elimination operator of exact differential order $r_N(q)$, and exact overlaps should appear as order drops in noncommutative elimination.

12.6 Inverse Fabius samples

Conjecture 12.5 (Non-P-recursiveness of inverse endpoint samples). *For every fixed $c > 0$, the sequence*

$$b_n(c) := F^{-1}(c2^{-n}) \quad (12.4)$$

for sufficiently large n is not P-recursive.

Unlike $F(c2^{-n})$, the inverse sequence decays only stretched-exponentially in $\sqrt{n}$ at leading order, a scale compatible with some P-recursive asymptotic templates. A proof will therefore need the complete endpoint structure, probably the nonconstant log-periodic factor in the corrected Lambert– W expansion, rather than a simple ratio argument. One possible route is to show that the asymptotic transseries violates the finite collection of exponential scales allowed by Birkhoff–Trjitzinsky theory for polynomial recurrences.

12.7 Saddle analysis of the order-zero sample OGF

Problem 12.6 (Sharp growth of A_c). *Insert the all-orders endpoint expansion of $F(c2^{-n})$ into*

$$A_c(r) = \sum_{n \geq 0} F(c2^{-n})r^n \quad (12.5)$$

and derive a complete saddle expansion as $r \rightarrow +\infty$. Determine whether

$$\log A_c(r) = \frac{(\log r)^2}{2 \log 2} + \text{lower log-log terms} + \mathcal{Q}_c(\log_2 \log r) + o(1) \quad (12.6)$$

with a nonconstant periodic function $\mathcal{Q}_c$.

This would create a new duality between the endpoint asymptotics of F and the growth theory of an entire function of order zero. The saddle index is of size $\log_2 r$, while the endpoint carrier itself contains a Lambert– W scale; their interaction should produce a two-level asymptotic inversion.

12.8 Formalization targets

Several proofs are unusually suitable for Lean formalization.

1. Polynomial-exponential independence and the exact rank theorem can be reduced to finite-dimensional linear algebra plus polynomial injectivity of $D + c$.
2. The collision criterion is a finite equivalence on sign vectors and $\{-1, 0, 1\}$ coefficient lists.
3. Rank submultiplicativity and Fekete’s lemma give the entropy theorem.
4. The non- D -finiteness proof needs only the ordinary-point uniqueness theorem and the already formalized zero multiplicities.
5. The deep-sample obstruction is an elementary asymptotic contradiction from monotonicity and (10.2).

A formal development would be small relative to the existing Fabius corpus and would certify several genuinely global negative statements that are otherwise easy to overlook.

13 Conclusion

The finite and infinite sinc products sit on opposite sides of a sharp holonomic boundary. A finite product is an exponential polynomial after one rational gauge, but its exact differential rank is not merely the number of factors: it is the signed support size of a Bernoulli convolution, equivalently the number of surviving top-order spline knots. This turns exact overlaps into minimal-ODE rank defects and produces a submultiplicative entropy that detects every restricted-coefficient algebraic relation.

At the dyadic parameter, the absence of overlaps exposes the full 2^N -dimensional structure. The odd frequency lattice yields a complete minimal operator, Thue–Morse coefficients, a Pochhammer factorization, Bell–Bernoulli coefficient formulae, and an exact Barnes– G discriminant. The infinite limit retains a first-order dilation equation but loses all finite linear differential description. Its zero

multiplicities, arithmetic residues, and singularity pattern force non-D-finiteness, trigonometric-field escape, and non-P-recursiveness of several natural moment and endpoint sequences.

The strongest open bridge is now clear: combine exact-overlap automata, parameterized difference Galois theory, and the repository’s Lambert– W endpoint transseries. That program could upgrade the present linear non-holonomicity to differential transcendence, solve the golden rank recurrence, and connect rank entropy to the much deeper entropy theory of Bernoulli convolutions.

A A compact theorem dependency map

Result	Inputs
Exact finite rank, Theorem 3.3	Exponential expansion (3.2); polynomial-exponential independence, Theorem 3.1 .
Rank–knot duality, Theorem 3.5	Distributional derivative of a box; exact finite rank.
Full-rank criterion, Theorem 4.1	Difference of two sign words is $2qP(q)$ with $P \in \{-1, 0, 1\}[x]$.
Rank entropy, Theorem 4.5	Block factorization (4.7); Fekete’s lemma.
Dyadic minimal ODE, Theorem 5.1	Odd-lattice/Thue–Morse parametrization; exact finite rank.
Bell–Bernoulli coefficients, Theorem 5.3	Newton/Bell transform and Bernoulli polynomial power sums.
Infinite non-D-finiteness, Theorem 8.2	Existing zero divisor; ordinary-point zero-multiplicity bound.
Field escape, Theorem 8.6	Residues $1 + \nu_2(m)$; rational Laurent expansion at $w = 1$.
Moment non-P-recursiveness, Theorem 9.2	Infinite non-D-finiteness; finite singular locus; Stanley dictionary.
Deep sample obstruction, Theorem 10.2	Integrated Fabius equation; monotonicity; exponential ratio separation.

B Selected exact computational certificates

The archive contains the complete files; this appendix records the smallest human-checkable certificates.

B.1 Golden level three

Using $q^2 = 1 - q$, the two zero-frequency words are

$$(+, -, -) : q - q^2 - q^3 = 0, \quad (-, +, +) : -q + q^2 + q^3 = 0,$$

with parity weights $+1$ and -1 . The other words reduce to the frequencies $\pm 2q, \pm 2q^2, \pm 2q^3$.

B.2 Dyadic level three

The positive frequencies are $1/8, 3/8, 5/8, 7/8$, so

$$\begin{aligned} \mathcal{P}_3(D) &= \left(D^2 + \frac{1}{64}\right) \left(D^2 + \frac{9}{64}\right) \left(D^2 + \frac{25}{64}\right) \left(D^2 + \frac{49}{64}\right) \\ &= D^8 + \frac{21}{16}D^6 + \frac{987}{2048}D^4 + \frac{3229}{65536}D^2 + \frac{11025}{16777216}. \end{aligned}$$

B.3 First cumulants

From (9.3),

$$\kappa_2 = \frac{1}{9}, \quad \kappa_4 = -\frac{2}{225}, \quad \kappa_6 = \frac{16}{3969}, \quad \kappa_8 = -\frac{16}{3825}. \quad (\text{B.1})$$

For example, $\mu_2 = \kappa_2 = 1/9$ and $\mu_4 = \kappa_4 + 3\kappa_2^2 = 19/675$.

C Reproducibility manifest

The ZIP archive contains:

<code>fabius_holonomic_frontiers.tex</code>	Complete LaTeX source of this report.
<code>fabius_holonomic_frontiers.pdf</code>	Compiled report.
<code>frontier_experiments.py</code>	Commented exact-arithmetic and high-precision experiment script.
<code>requirements.txt</code>	Python dependencies.
<code>CORPUS_AUDIT.md</code>	Repository snapshot, search method, and exclusion boundaries.
<code>NUMERICAL_README.md</code>	Reproduction instructions and status caveats.
<code>data/</code>	Exact rank tables, operator coefficients, residuals, discriminants, separation tables, and vector/raster figures.

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